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Date - 06 July 2026

THINGS I LERANED

Sustainable Development Goals (SDGs)

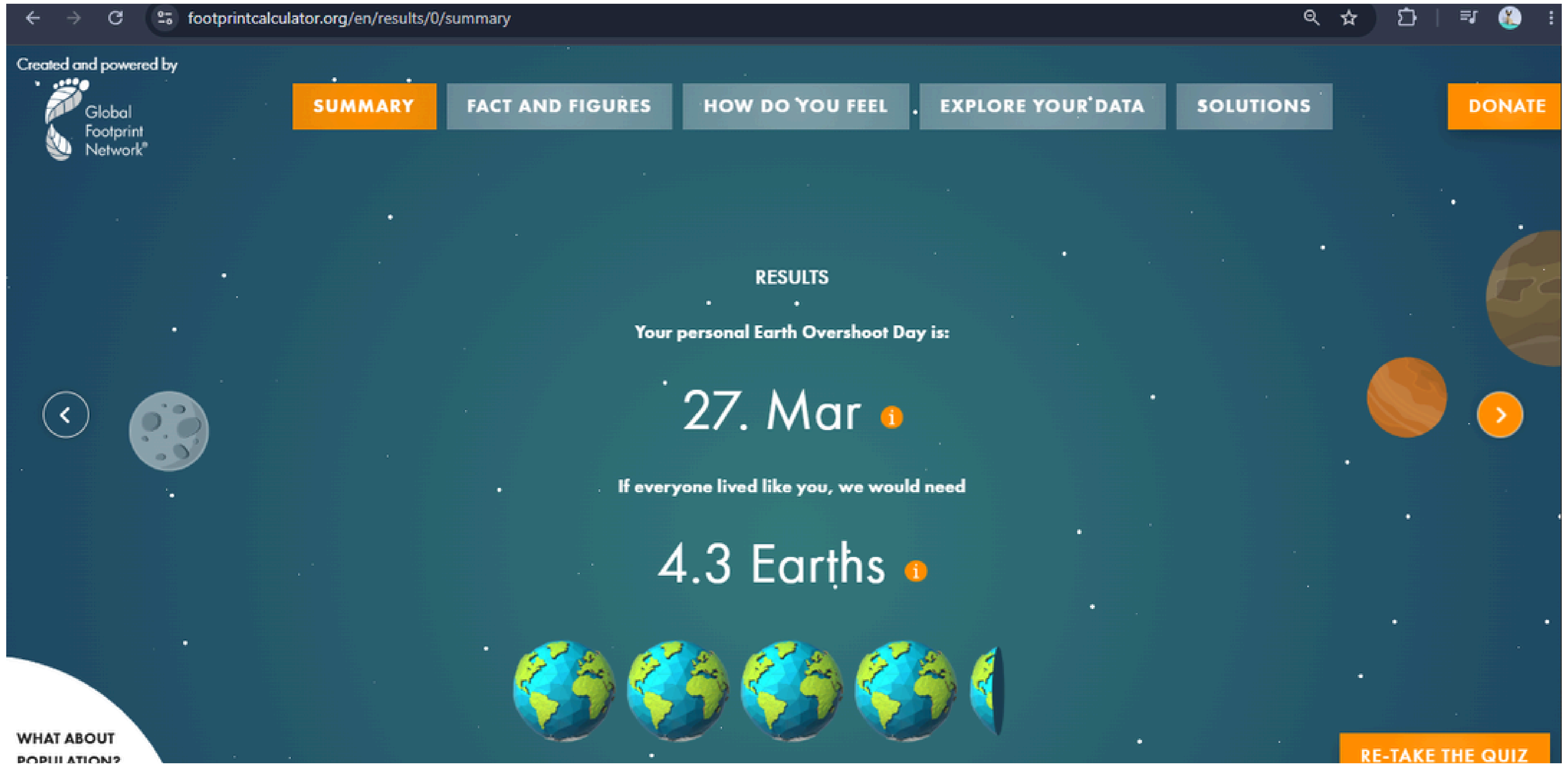
- The collection of 17 SDGs works toward achieving sustainable development within the 2030 timeframe.
- The SDG I find most compelling: SDG 17 – Partnerships for the Goals
- Promotes cooperative efforts among governmental bodies, business sectors, and private citizens.
- Fosters creative solutions and worldwide environmental responsibility.

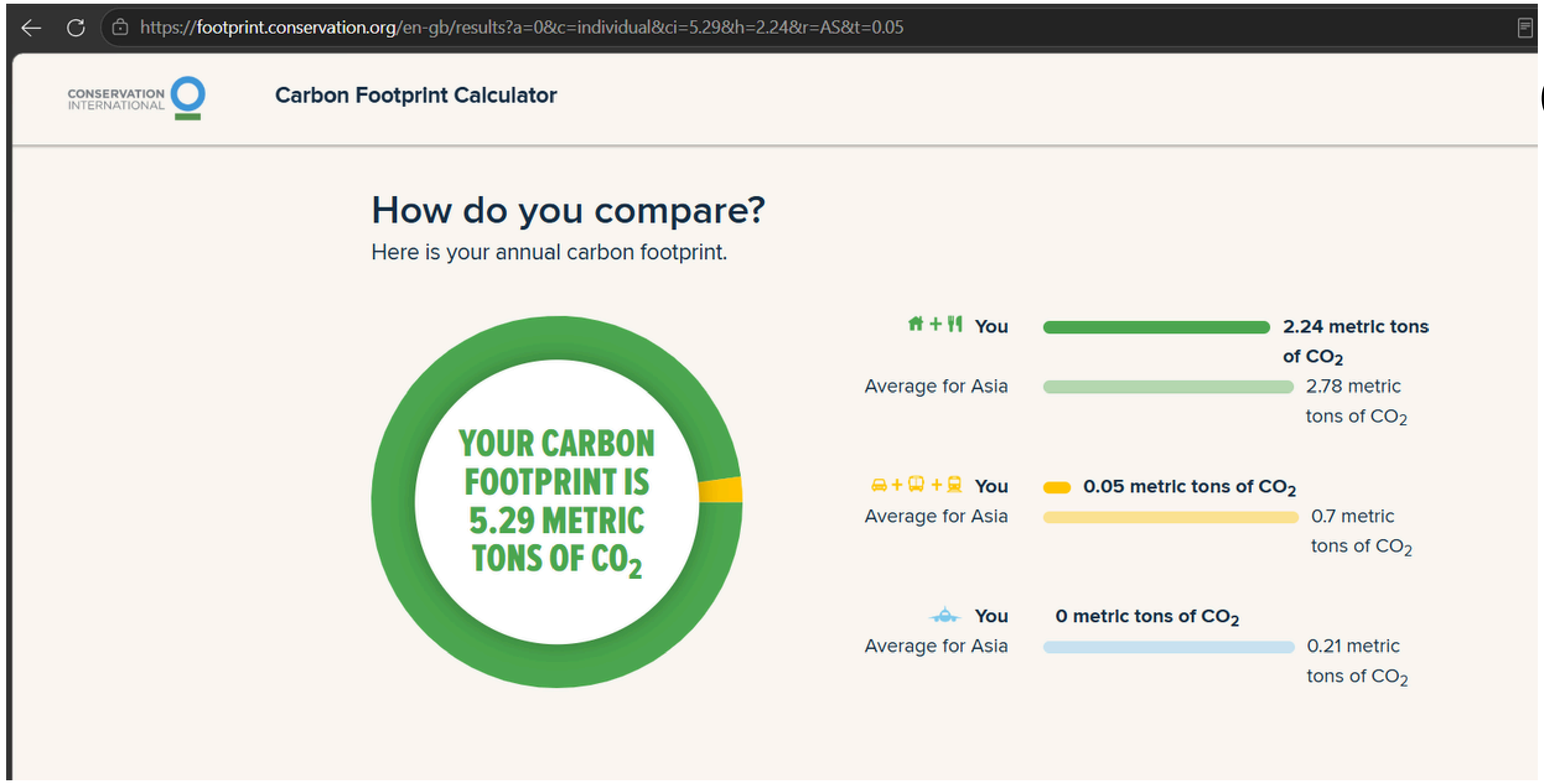
AI Data Center Documentary

- Data centers supporting AI operations require substantial quantities of electrical power and water resources.
- The expansion of AI technology leads to heightened energy consumption and greenhouse gas output.
- Environmentally conscious AI approaches encompass:
  - Sustainable Energy Sources
  - Advanced Temperature Control Methods
  - Power Usage Enhancement

Ecological Footprint Calculator

- Personal ecological footprint



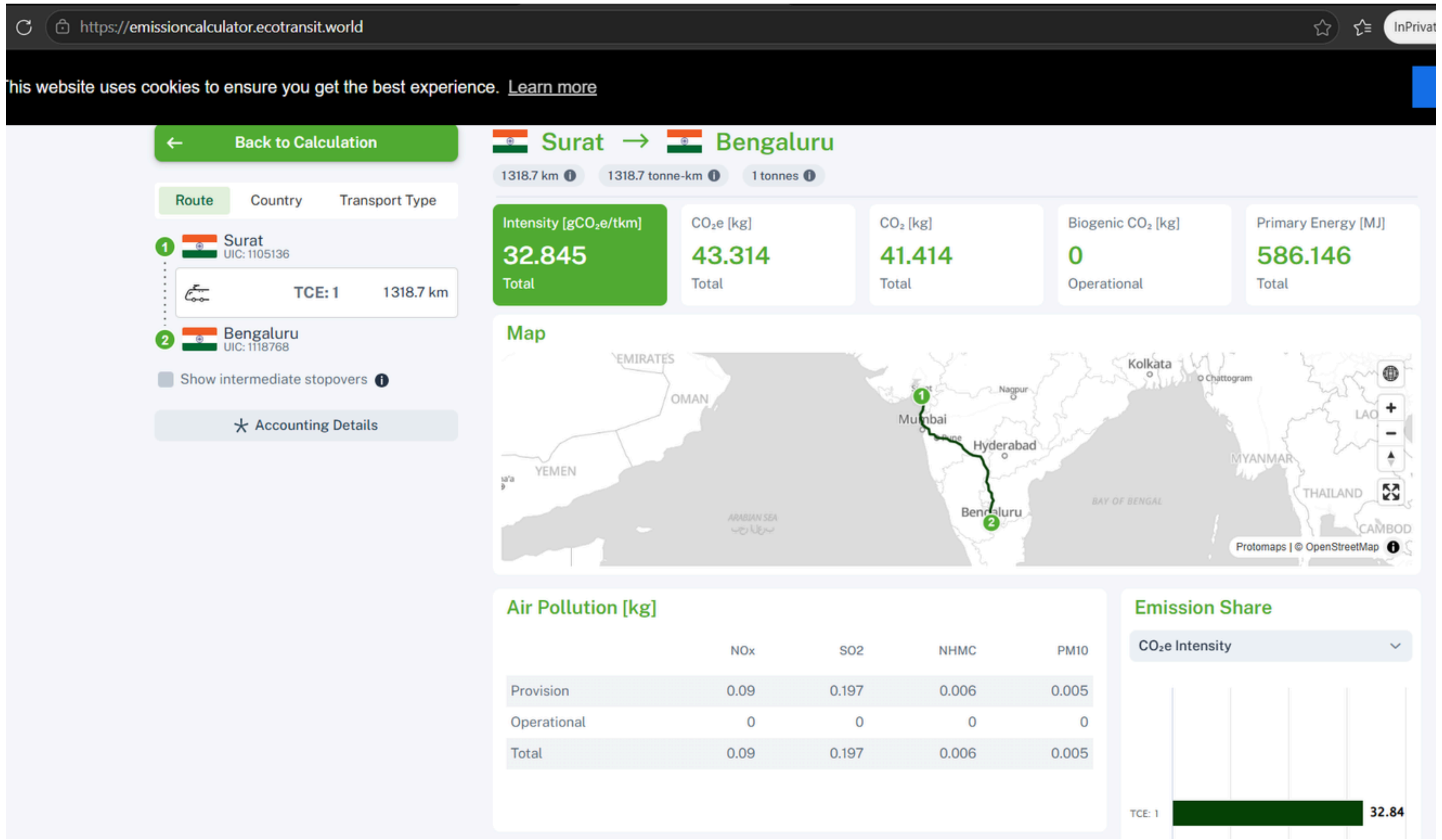


Carbon Footprint Calculator

- Household emissions
- Transportation emissions

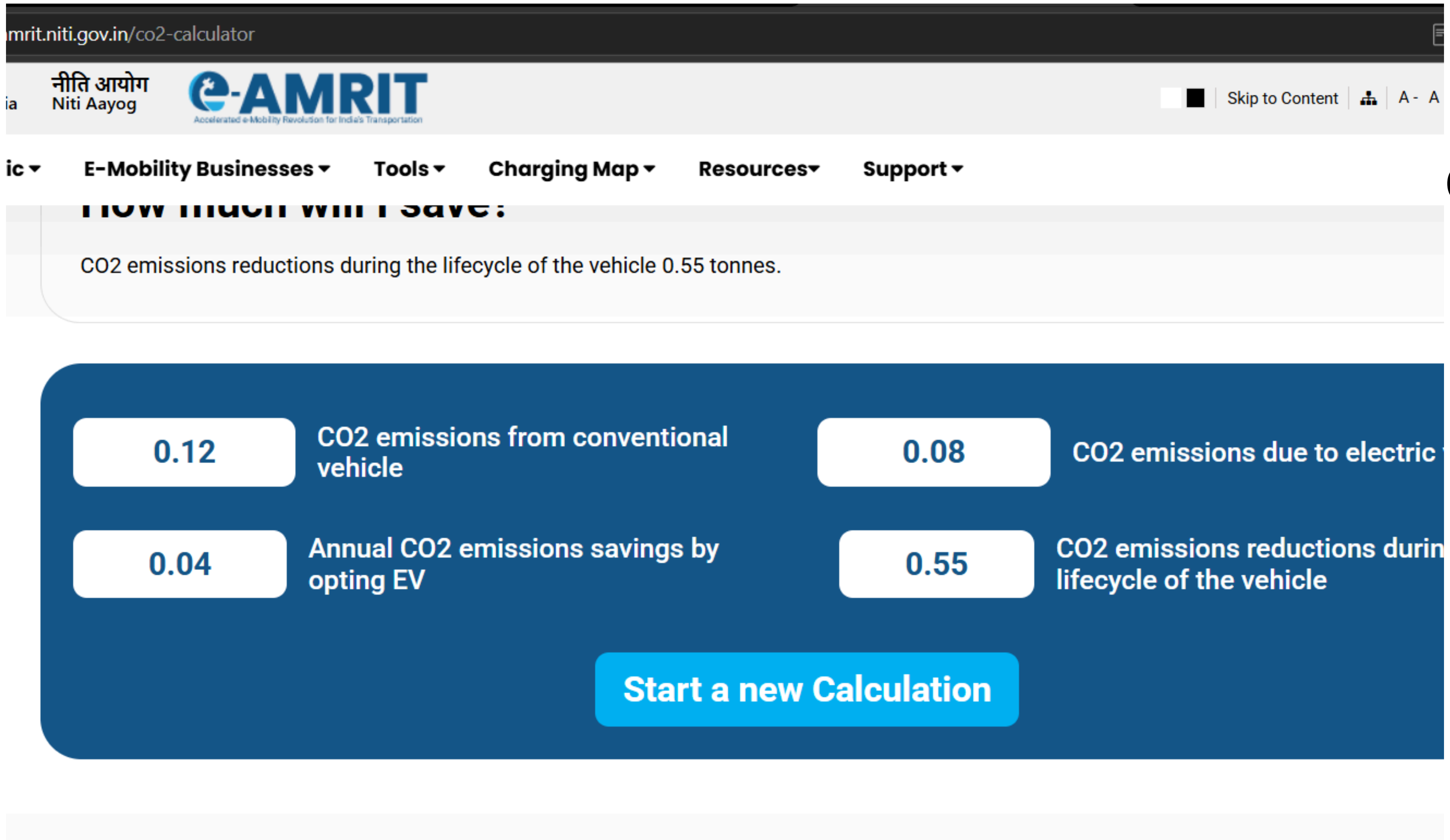
EcoTransit CO_2 Calculator

- Comparison of transport emissions



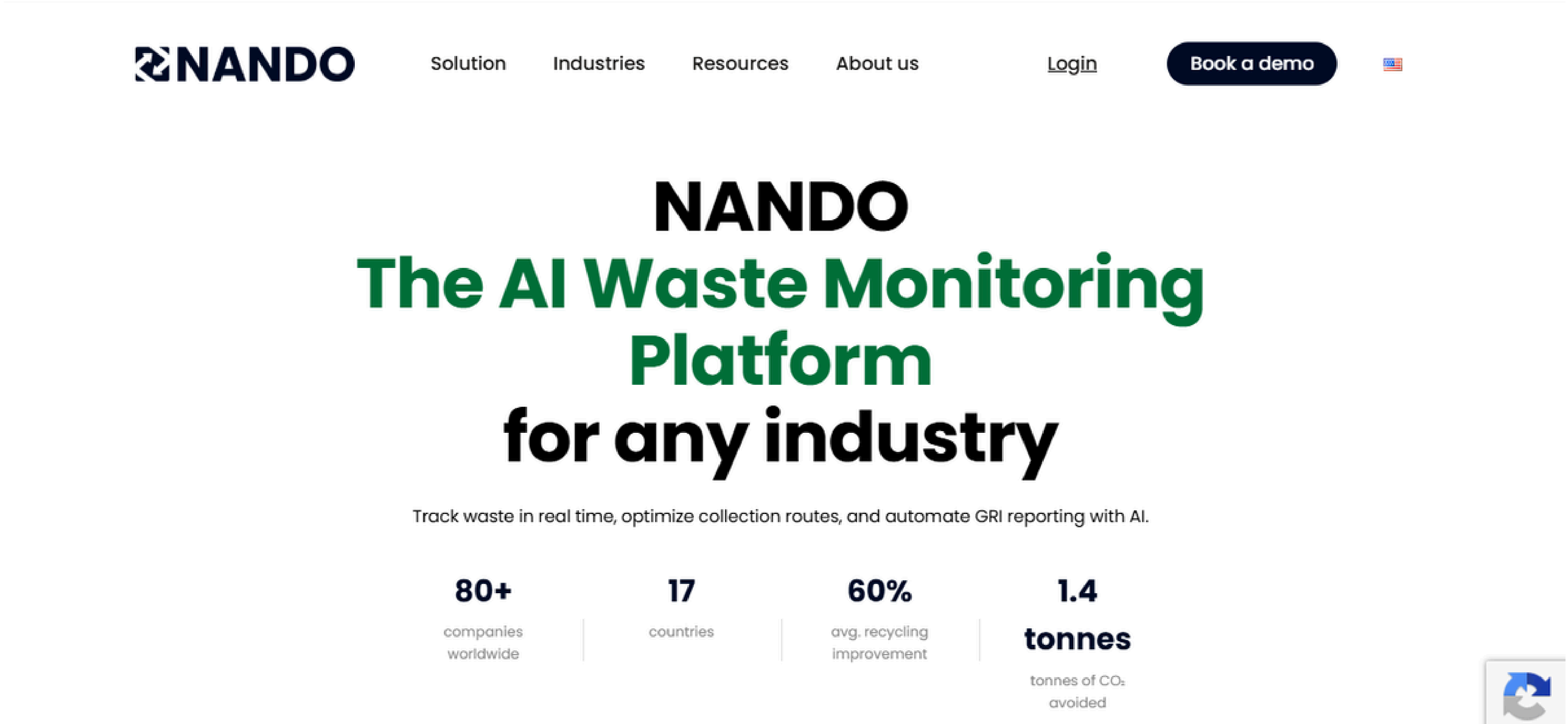
e-AMRIT CO_2 Calculator

- Vehicle emission analysis



En-ROADS Climate Simulator

- Climate policy simulation



NANDO AI

- AI-powered waste monitoring

Green Skilling, Sustainability & AI

Green Skilling

- Builds expertise in environmentally responsible practices and hands-on capabilities.
- Generates job opportunities within sustainable sectors.
- Encourages the adoption of earth-conscious methods.

Sustainability

- Harmonizes social well-being, environmental health, and economic growth.
- Preserves Earth's natural assets.
- Enables ongoing ecological safeguarding for future generations.

AI for Sustainability

- Technology-driven farming techniques for optimal resource use
- Strategic power consumption management
- Assessment and tracking of greenhouse gas outputs

Challenges

- Excessive power requirements
- Significant water consumption for cooling facilities
- Accumulation of discarded electronic equipment
- Greenhouse gas production from AI computing infrastructure

CAPSTONE PROJECT - AI Campus Sustainability Dashboard

Problem Statement

- Educational institutions often lack a centralized system to monitor electricity usage, water consumption, waste generation, and sustainability performance.

Proposed Solution

- Develop a dashboard that tracks campus sustainability metrics and provides AI-driven recommendations for improvement.

AI Component

- Predictive analytics and anomaly detection.

Related SDGs: 11, 12, 13, 17